Architectural Issues
— Internal Model

(控制结构问题 — 内模设计)

Main Contents of Course (课程主要内容)

Mechanical (机械)

Electrical (电路)

Electro-Mechanical (机-电)

Fluid (流体)

Thermal (热系统)

Dynamic Modeling (动态建模)

Structural selection (模型结构)

System Identification (系统辨识)

Parameter estimation (参数估计)

Control System Design (控制系统设计)

Root-locus design (根轨迹设计)

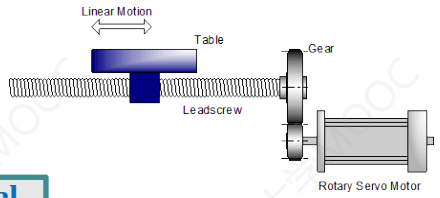
Pole placement design (极点配置)

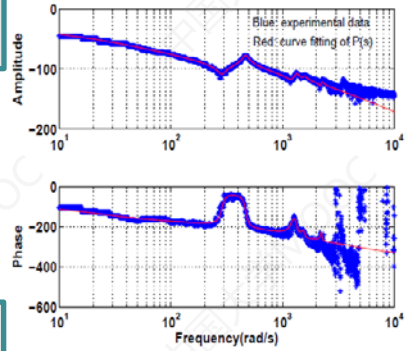
Frequency design (频域设计)

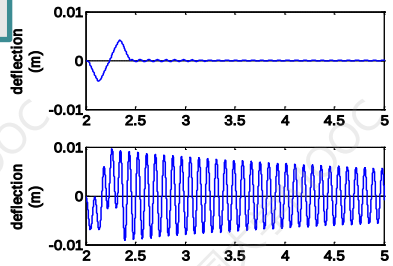
Architectural issues (控制结构)

Design limitations (性能局限)

State-feedback (状态反馈)







Models of Disturbances and References

- **Step Type Disturbances and References** (阶跃类型的干扰和参考输入)
  - **Can be modeled as free response of an LTI system of** (可以被建模为LTI系统的自由响应)

$$\dot{d}_i(t) = 0 \quad \text{or} \quad P_{dis}(s) = \frac{1}{s} \text{ with } u = 0$$

$$sD_{dis}(s) - d_i(0) = 0 \Rightarrow D_{dis}(s) = \frac{d_i(0)}{s} = d_i(0)P_{dis}(s)$$

- **Sinusoidal Type Disturbances and References** (正弦类型的干扰和参考输入)
  - **A sinusoidal disturbance** $d_i(t) = A_d \sin(\omega_d t + \varphi_d)$ **can be modeled as free response of an LTI system given by** (正弦干扰可以被建模为LTI系统的自由响应)

$$\ddot{d}_i(t) + \omega_d^2 d_i(t) = 0 \quad \text{or} \quad P_{dis}(s) = \frac{1}{s^2 + \omega_d^2} \text{ with } u = 0$$

$$\dot{d}_i = A_d \omega_d \cos(\omega_d t + \varphi_d)$$

$$\ddot{d}_i = -\omega_d{}^2 \underbrace{A_d \sin(\omega_d t + \varphi_d)}_{d_i(t)}$$

$$s^2 D_{dis}(s) - sd_i(0) - \dot{d}_i(0) + \omega_d^2 D_{dis}(s) = 0$$

$$\Rightarrow D_{dis}(s) = \frac{sd_i(0) + \dot{d}_i(0)}{s^2 + \omega_d^2} = P_{dis}(s)[sd_i(0) + \dot{d}_i(0)]$$

poles at $\pm \omega_d j$

Models of Disturbances and References

- **Mixed Step and Sinusoidal Type Disturbances and References (混合阶跃和正弦类型的干扰和参考输入)**
  - **can be modeled as free response of an LTI system given by (可以被建模为LTI系统的自由响应)**

$$d_i(t) = A_{d0} + A_d \sin(\omega_d t + \varphi_d)$$

$$\dddot{d}_i(t) + \omega_d{}^2 \dot{d}_i(t) = 0 \quad \text{or} \quad P_{dis}(s) = \frac{1}{s(s^2 + \omega_d{}^2)} \text{ with } u = 0$$

$$d_i(t) = A_1 \sin(\omega_1 t + \varphi_1) + A_2 \sin(\omega_2 t + \varphi_2)$$

$$P_{dis}(s) = \frac{1}{(s^2 + \omega_1{}^2)} \frac{1}{(s^2 + \omega_2{}^2)}$$

Models of Disturbances and References

- **General *Bounded* Disturbances and References (一般有界干扰和参考输入)**
  - **Assume to be modeled as free response of an LTI system given by (假设被建模为LTI系统的自由响应)**

$$d_i^{(q)}(t) + \gamma_{q-1} d_i^{(q-1)}(t) + \ldots + \gamma_0 d_i(t) = 0$$

or

$$\Longrightarrow \quad P_{dis}(s) = \frac{1}{\Gamma_{dis}(s)}, \qquad \Gamma_{dis}(s) = s^q + \gamma_{q-1} s^{q-1} + \ldots + \gamma_0$$

Disturbance generating polynomial (干扰生成多项式)

$$d_i(s) = \frac{N_{dis}(s)}{\Gamma_{dis}(s)}$$

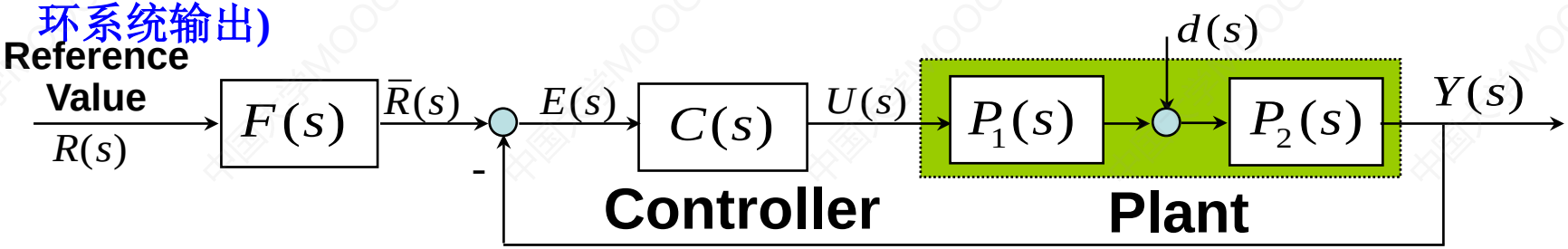
order of $N_{dis}(s) \le q-1$

Q: What is the generating polynomial for disturbances with frequencies at $\omega_{d1},\ldots,\omega_{dm}$? (频率为$\omega_{d1},\ldots,\omega_{dm}$干扰的生成多项式是什么)

$$\Gamma_{dis}(s) = (s^2 + \omega_{d1}^2)\ldots(s^2 + \omega_{dm}^2) \quad \text{of order of } 2m$$

Internal Model Principle for Disturbances

- **Closed-Loop System Output due to Disturbance** (因干扰引起的闭环系统输出)



$$Y(s) = S(s)P_2(s)d(s) = \frac{D_{p1}(s)\cancel{D_{p2}(s)}D_c(s)}{A_{CL}(s)}\frac{N_{p2}(s)}{\cancel{D_{p2}(s)}}\frac{N_{dis}(s)}{\Gamma_{dis}(s)}$$

$$sY(s) = s\frac{D_{p1}D_{p2}D_c}{A_{CL}(s)}\cdot\frac{N_{p2}}{D_{p2}}\cdot\frac{N_{dis}}{\Gamma_{dis}} = \frac{sD_{p1}D_c}{A_{CL}(s)}\frac{N_{p2}N_{dis}}{\Gamma_{dis}}$$

$\Rightarrow sY(s)$ has all poles (i.e., poles of $A_{CL}(s)$ or all CL poles) in LHP,

$$\Rightarrow y(t) \to y_\infty = \lim_{s\to 0} sY(s) = 0 \times \frac{D_{p1}(s)\bar{D}_c(s)N_{p2}(s)N_{dis}(s)}{A_{CL}(s)} = 0$$

- **Condition for *Zero Steady-State Output* due to Disturbances** (对干扰零稳态输出的条件)

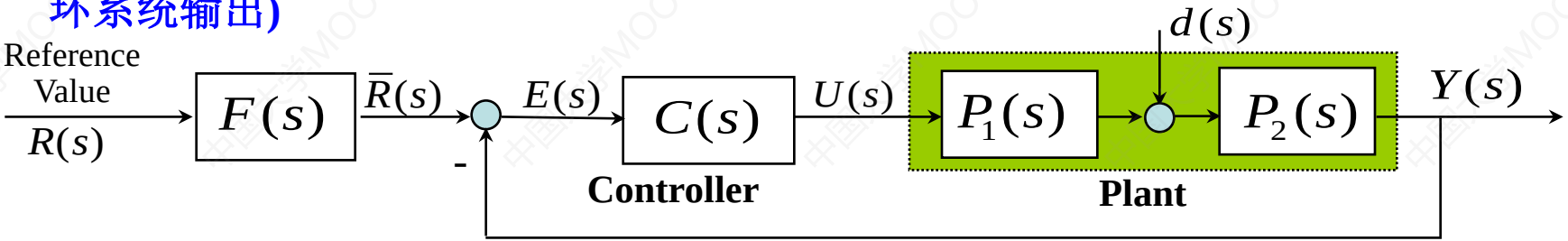
$$D_c(s) = \Gamma_{dis}\bar{D}_c(s)$$

Internal Model Principle for Disturbances

- **Internal Model Principle (内模原理)**
  - ✓ ***Steady-state disturbance compensation*** **requires that generating polynomial of disturbances be included as part of the controller denominator, which is called the *Internal Model Principle*. (稳态干扰补偿要求干扰生成多项式作为控制器分母的一部分，这称为内模原理)**
  - ✓ Note that the roots of the generating polynomial, in particular, the ones on the imaginary axis, impose the same performance trade-offs on the closed-loop as of those poles were part of the plant ! **(注意生成多项式的根特别是虚轴上的根是系统极点的一部分，同样存在性能平衡问题)**
- **Industrial Applications (工业应用)**
  - Hard Disk Drive Industry for compensation of repeatable run-outs (i.e., periodic disturbances and references due to the constant spinning speed of disks) **(硬盘驱动系统的重复运动补偿(即由于磁盘的恒定旋转速度而引起的周期性干扰和参考输入))**
  - Roll-eccentricity compensation in Rolling Mills, …. **(轧机轧辊偏心补偿)**

Internal Model Principle for Reference Tracking

- **CL System Output Error due to References** (对参考输入引起的闭环系统输出)



$$Y(s) = T(s)F(s)R(s)$$

$$E(s) = \bar{R}(s) - Y(s) = S(s)F(s)R(s) = \frac{D_p(s)D_c(s)}{A_{CL}(s)} \frac{N_F(s)}{D_F(s)} \frac{N_r(s)}{\cancel{\Gamma_r(s)}}$$

$\cancel{\Gamma_r(s)}\bar{D}_c(s)$

- **Condition for *Zero Steady-State Error* due to References** (对参考输入零稳态误差的条件)

$$D_c(s) = \Gamma_r(s)\bar{D}_c(s)$$

Summary

- Internal Model Principle for Perfect Disturbance Rejection at Steady-State（稳定状态下干扰抑制的内模原理）
  - Models for Deterministic Disturbances and References (确定性干扰和参考的模型)
  - Internal Model Principle for disturbance rejection (干扰抑制的内部模型原理)
  - Internal Model Principle for reference tracking (参考跟踪的内部模型原理)